

Controlling deltamethrin-resistant *Rhipicephalus microplus* with a phytoformulation of *Annona muricata* and *Piper longum*

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ARTICLE INFO

Keywords:

Tick
AIT
LIT
Soxalation
HP-TLC
Phytoformulation

ABSTRACT

Ticks are one of the most significant vectors for transmitting diseases in livestock. Synthetic acaricides are the primary method for controlling tick populations; however, their indiscriminate application has led to widespread acaricide resistance and environmental concerns. In response to these challenges, this research has focused on developing eco-friendly herbal acaricides that ensure animal safety while minimizing environmental contamination and residual toxicity. Plants produce secondary metabolites, some of which possess insecticidal properties and have potential as natural acaricides. In the present study, the acaricidal properties of five plant extracts were evaluated against *Rhipicephalus microplus*. The hexane extracts of *Annona muricata* leaves and *Piper longum* seeds demonstrated significant acaricidal activity, with IC_{50} values of 0.34 % and 0.184 % in larval immersion tests and 1.39 % and 0.71 % in adult immersion tests, respectively. A stable phytoformulation was developed incorporating these extracts, and its physicochemical properties were monitored over 12 months to ensure consistency and quality. HPTLC profiling identified piperine and α -caryophyllene as marker compounds for quality control. *In vitro* assays using phytoformulation showed lower IC_{50} values than individual plant extracts in deltamethrin-resistant ticks at the larval stage. Also, at the adult stage, the phytoformulation inhibited egg hatchability and disrupted ovarian structures. Toxicity studies confirmed the phytoformulation's safety for animal use. Field trials on natural tick-infested cattle demonstrated a mean efficacy of 83.3 ± 3.5 % within 48 h of application. This study suggests that phytoformulation could serve as a sustainable, safe, and effective way to manage deltamethrin-resistant *Rhipicephalus microplus* infestations.

1. Introduction

Ticks are hematophagous ectoparasites of immense veterinary and medical importance, affecting livestock, wildlife, and humans globally. Their blood-feeding behavior causes anemia, reduced productivity in milk and meat, decreased body weight, and damaged hide quality, and they also act as vectors for numerous pathogens. Tick and tick-borne diseases (TTBDs) are estimated to cause global economic losses ranging US\$22–30 billion annually (Singh et al., 2022).

Rhipicephalus microplus is an economically significant ectoparasite that affects livestock in tropical and subtropical regions. It feeds on a single host, primarily on cattle to complete the parasitic stages of its life

cycle: larva, nymph, and adult. *R. microplus* is a vector for various pathogens, including *Babesia bovis*, *Babesia bigemina*, and *Anaplasma marginale*, (Suarez and Noh, 2011). The primary method for tick control is the usage of chemical acaricides, including macrocyclic lactones, formamidines, organophosphates, and pyrethroids. However, extensive use of these acaricides has led to several challenges, such as environmental contamination due to their persistence in soil and water, contamination of livestock products, and the accelerated development of acaricide-resistant tick populations (Abbas et al., 2014, Dzemo et al., 2022). An anti-tick vaccine is an alternative strategy, but vaccines developed to date offer limited cross-protective efficacy (Parizi et al., 2012). These issues highlight the need for more effective and sustainable

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tick control solutions.

Plant-based acaricides, comprising single molecules and crude extracts, are emerging as a promising alternative (Quadros et al., 2020, Gonzaga et al., 2023). Furthermore, combining synthetic acaricides with these natural compounds can enhance effectiveness against tick infestations (Vaou et al., 2022). An additional advantage of plant-based acaricides is their ability to act synergistically. The complex mixture of secondary metabolites in plant extracts often works together to enhance the overall acaricidal effect. This synergistic action not only increases efficacy but also reduces resistance development. Numerous plants have been screened for their efficacy against ticks in *in vitro* conditions. However, only limited studies have taken these prepared phytoformulations and tested them *in vivo* under field conditions (Sharma et al., 2022).

This study tested five plant extracts, namely *Piper longum* (Long pepper), *Annona muricata* (Soursop), *Curculigo orchoides* (Golden eye grass), *Bauhinia variegata* (Mountain ebony), and *Azadirachta indica* (Neem), for their potential as acaricides. *Piper longum*, belonging to the Piperaceae family, is a common plant widely used as a spice and is known for its medicinal properties, including anti-inflammatory and anticancer activities (Mujumdar et al., 1990, Fujihashi et al., 2023). Additionally, it has been reported to possess insecticidal and acaricidal properties (Madhu et al., 2011, Singh et al., 2017). *Annona muricata*, belonging to the Annonaceae family, exhibits immune-modulatory, anti-inflammatory, and antimalarial activities (Gavamukulya et al., 2017) and has also been reported to possess mosquitocidal properties (Parthiban et al., 2020). *Curculigo orchoides* contains flavonoids, phenolics, and saponins, which contribute to its antioxidant, anti-inflammatory, and insecticidal properties (Nie et al., 2013). *Bauhinia variegata*, is rich in flavonoids, alkaloids, and tannins, and is recognized for its antimicrobial, antioxidant, and insecticidal activities (Mishra et al., 2013). *Azadirachta indica* is renowned for its potent bioactive limonoids, such as azadirachtin and nimbin, which exhibit strong insecticidal and growth-regulating effects against a broad range of pests (Wylie and Merrell, 2022).

This study focuses on the *in vitro* evaluation of plant extracts against larval and adult stages of *Rhipicephalus microplus*. The active extracts were selected to prepare a phytoformulation, and the effects of phytoformulation on deltamethrin-resistant ticks were assessed under *in vitro* conditions. Additionally, the cytotoxicity and *in vivo* efficacy of the phytoformulation were evaluated.

2. Materials and methods

2.1. Ethical approval

The procedures for animal experiments were approved by the Institutional Animal Ethics Committee (IAEC) with approval no. IAEC/NIAB/2023/29/AS and performed in accordance with the guidelines from the Committee for Control and Supervision of Experiments on Animals (CCSEA).

2.2. Plant crude extract preparation

All plant materials were procured from Bhagyashree Herbal Farms, a

certified organic wild herbal supplier (20 ° 42' 35.9964" N. 81 ° 33' 0.0108" E). Plant parts used for this study are mentioned in Table 1. A hundred grams of each dried plant material was ground into a coarse powder and subjected to a 48 h soxhlation extraction process in increasing order of solvent polarity from hexane, chloroform, and methanol. Each extract was concentrated using a rotary evaporator and stored at 4 °C.

2.3. Collection of *Rhipicephalus microplus* ticks

Engorged female *R. microplus* ticks were collected from unorganised cattle farms, located in the Parbhani region of Maharashtra, India (19 °15'33"N 76 °46'59"E). The ticks were cleaned with 30 % ethanol (Sharma et al., 2012) and used for the adult immersion test (AIT). Few ticks were placed in glass vials covered with muslin fabric for egg-laying. The tick-containing vials were maintained in a BOD chamber at 27 ± 2 °C and relative humidity of 80 ± 5 %. The eggs were collected following the oviposition period (12–14 days) and stored in separate glass vials to allow larval hatching. After 8–15 days of hatching, the larvae were used for the larval immersion test (LIT). All *in vitro* studies were conducted at NIAB, Hyderabad, India, while the *in vivo* studies were carried out in the Parbhani region of Maharashtra, India.

2.4. Adult immersion test (AIT)

The AIT was performed as described previously (Drummond et al., 1973) with slight modifications. Five engorged female ticks were immersed in a 5 % plant extract solution prepared in 30 % ethanol and 0.2 % Tween-20, and incubated for 5 min at room temperature. After incubation, the ticks were dried using tissue paper and placed in glass vials covered with muslin cloth. The vials were maintained in a BOD chamber at 27 ± 2 °C and relative humidity of 80 ± 5 %. Each concentration was tested in five replicates. Ticks treated with 30 % ethanol and 0.2 % Tween-20 served as the control group (Sharma et al., 2012). The mortality rate was recorded 48 h post-treatment, and surviving ticks were observed for the oviposition over 14-day period.

2.5. Larval immersion test (LIT)

The LIT was performed as described previously (Shaw, 1966) with slight modifications. A total of 60–100 larvae were transferred to a 1.5 mL centrifuge tube containing 300 µL of either 1.5 % hexane, 5 % chloroform or methanol plant extract solution prepared in 30 % ethanol and 0.2 % Tween-20. Each tube was shaken vigorously and incubated for 5 min at RT. Following incubation, the solution was pipetted out, and the tube was dried by placing a filter paper strip inside to absorb any residual solution. The tubes were then covered with muslin cloth and placed in a BOD incubator maintained at 27 ± 2 °C and 80 ± 5 % relative humidity. Larvae treated with 30 % ethanol and 0.2 % Tween-20 served as the control group (Sharma et al., 2012). After 24 h, the number of dead and live larvae were counted to calculate the mortality rate. Each experiment was performed thrice in triplicate.

The IC₅₀ values for the active plant extracts were calculated across a concentration range of 5 % to 0.1 % using dose-response inhibition analysis in GraphPad Prism software.

Table 1

List of selected plants, their parts utilized, and the corresponding percentage yield obtained from 100 g of plant material.

S. No	Botanical Name	Family	Part used	Hexane Yield (grams)	Chloroform	Methanol
1	<i>Curculigo orchoides</i>	Hypoxidaceae	Rhizomes	3.5	4.6	16.36
2	<i>Bauhinia variegata</i>	Fabaceae	Bark	0.92	0.23	25.05
3	<i>Azadirachta indica</i>	Meliaceae	Leaves	4.49	6.5	18.5
4	<i>Annona muricata</i>	Annonaceae	Leaves	4.26	2.8	15.6
5	<i>Piper longum</i>	Piperaceae	Fruits	2.15	4.25	5.7

2.6. Preliminary-phytochemicals screening

The plant extracts which showed promising acaricidal activity, specifically the hexane extracts of *A. muricata* and *P. longum*, were screened for phytochemical screening to identify the presence of carbohydrates, amino acids, phenols, glycosides, steroids, terpenes, and saponins, using methods described in supplementary method file 1.

2.7. Preparation of phytoformulation and physicochemical properties

For preparing phytoformulation, 1.5 g of each *A. muricata* and *P. longum* hexane extracts and 1 mL of *C. citriodora* oil, were added to the final volume of 100 mL shampoo base. All the components were mixed manually. The prepared phytoformulation was evaluated for various physicochemical properties, including color, pH, percentage of solid contents, surface tension, foaming ability, stability, and wetting time using methods described in supplementary method file 1.

2.8. HPTLC analysis of the selected plant extract

HPTLC analysis of active extracts was performed on 10 cm × 10 cm TLC aluminum plates pre-coated with 20 mm layer thickness of silica gel 60F₂₅₄ (Merck KGaA, 64271 Darmstadt, Germany) according to standard protocol. Sample of hexane extract of *A. muricata*, and active marker α -Caryophyllene were applied as 6 mm band width using Camag 100 μ L sample syringe (Hamilton, Switzerland) with a Linomat 5 applicator (Camag, Switzerland) under a flow of N₂ gas. The chamber was saturated with mobile phase vapor (hexane, ethyl acetate 98: 2) for 25 min at RT, and plates were kept at a distance of approximately 80 mm from the point of application. After separation, the plates were air-dried, derivatized with para-anisaldehyde solution, and analyzed using a Camag TLC Scanner 3 (slit dimensions: 4 mm × 0.45 mm, scanning speed: 100 mm/s), operated via winCATS software (Version 3.2.1). Similarly, for hexane extract of *P. longum* and active marker piperine, separation was achieved using a mobile phase consisting of toluene: ethyl acetate: diethyl ether (60:30:10), the separated components were visualized under UV absorbance at 254 nm.

2.9. Determination of acaricide resistance ratio

Further, the ticks were classified as acaricide-resistant or sensitive based on the assay mentioned. LIT was performed using different concentrations of acaricides, such as Ivermectin, Deltamethrin, and Coumaphos, to test the resistance level in ticks collected from the field. The IC₅₀ values for the deltamethrin resistance in ticks were calculated, and the resistance ratio was determined using the reference report (Sharma et al., 2012).

Resistance ratio = IC₅₀ values of the field tick isolate/IC₅₀ values of the reference susceptible isolate.

2.10. Preparation of larval whole-body homogenate

The larvae of *R. microplus* (10 mg) was treated at an IC₅₀ value of deltamethrin was taken in a 1.5 mL centrifuge tube, immersed in liquid nitrogen, and crushed using a mortar pestle. Ice-cold PBS (pH 7.2) was added to the crushed samples and sonicated for 5 min (5 s ON and 10 sec OFF cycle at 30 % amps). The larval whole-body homogenates were centrifuged at 13,800xg at 4 °C for 15 min. The clear supernatant was collected, and the total protein concentration was estimated using Pierce™ BCA Protein Assay Kit according to the manufacturer protocol. The supernatant was then used to investigate the changes in biochemical constituents, Glutathione S-transferase (GST), and Acetylcholinesterase (AChE) activity. The untreated larvae homogenate was used as a control.

2.11. Glutathione S-transferase (GST) assay

Glutathione S-transferase activity in the larvae extract of *R. microplus* was performed as previously reported (Habig and Jakoby, 1981). Briefly, 10 μ L of the homogenate was added to 200 μ L of the reaction mixture (10 mM reduced glutathione and 3 mM CDNB in 1 M phosphate buffer, pH 6.5) and incubated for 20 min at RT. After incubation, the absorbance was measured at 340 nm using the multimode reader. The enzyme activity was reported as μ M of conjugate produced/min/mg protein, using the extinction coefficient (13600 M⁻¹ cm⁻¹). Each experiment was performed thrice in duplicates.

2.12. Acetylcholinesterase (AChE) assay

The AChE activity in larvae extract was performed as previously reported with slight modification (Ellman et al., 1961). Briefly, 20 μ L of larvae homogenate of *R. microplus* was added to 200 μ L of the reaction solution (1.25 mM acetylthiocholine iodide (ACT), 1 mM 5, 5'-dithio-bis-2-nitrobenzoic acid in sodium phosphate buffer, pH 7) and incubated for 5 min at RT. After incubation, the absorbance was measured at 405 nm in the multimode reader. AChE activity was expressed in μ M acetylthiocholine hydrolyzed/min/mg of protein. The experiment was performed thrice in duplicate.

2.13. In vitro efficacy of phytoformulation on deltamethrin-resistant larvae and adults of *Rhipicephalus microplus*

Deltamethrin-resistant ticks were used for the larval immersion test at concentrations of 0.05 %, 0.025 %, 0.0125 %, 0.0061 %, 0.0031 %, and 0.0015 %, and for the adult immersion test, at concentrations of 100 %, 50 %, 25 %, and 12.5 % and 6.12 % of the phytoformulation.

2.14. Stereo microscopic and scanning electron microscopy examination of ovary and eggs

After treatment with the IC₅₀ values of formulation (concentration of *A. muricata* and *P. longum*=0.046 %), adult ticks were dissected on the 5th day, and their ovaries were removed. Stereo microscopy (10× magnification) was used to examine the morphological differences between the ovaries of treated and control ticks. Eggs were collected from ticks treated with formulation at IC₂₅ value (concentration of *A. muricata* and *P. longum*=0.023 %) and examined under stereo microscope and scanning electron microscopy (SEM) along with their respective control. For SEM analysis, eggs were fixed in 2.5 % glutaraldehyde in PBS (pH 7.2) for 2 h. Following fixation, samples were rinsed thrice in PBS (pH 7.2) and stained with 2 % uranyl acetate for 1 h. Samples were again washed thrice in PBS (pH 7.2), and dehydrated in ethanol solutions at concentrations of 25 %, 50 %, 75 %, 90 %, and 100 % for 10 min each. Finally, the eggs were mounted on silver stubs, gold-coated by cathodic spraying (Polaron gold), and observed under a scanning electron microscope (SEM Voltage: 5kV, Carl Zeiss-EV018).

2.15. Safety study

The formulation was tested for safety study in a rabbit model following OECD guidelines (410, 404). The animals ($n = 9$, 3 per group) of groups I, II, and III were given topical application with formulation 3×3 cm, respectively. The animals ($n = 6$, 3 per group) in groups IV and V were treated with shampoo base, respectively. The skin was observed before and 24 h post-application. Any visible changes at the application site of the formulation, like staining, pruritis, and erythematic lesions, were recorded at different time intervals before and after treatment with respective control.

2.16. In vivo field trials

The field trials for the phytoformulation were conducted following the WAAVP guidelines (Holdsworth et al., 2022), with minor modifications. The trials were carried out on naturally tick infested animals from unorganized farms in the Parbhani region of Maharashtra, India. Data were collected during the peak tick infestation period between February and April. The animals ($n = 36$, 6 per group) in Groups I to IV were treated with the formulation, while Group V and VI was treated with the shampoo base. Tick infestation levels were assessed by counting the number of ticks in a 6×6 cm patch on the neck, thigh, and ear areas for each animal. The approximate mean number of ticks per animal was calculated before treatment and two days post-treatment. The acaricidal effect of phytoformulation was evidenced by photographs.

2.17. Statistical analysis

All statistical analyses were performed using GraphPad Prism 8.0 (GraphPad Software, San Diego, California, USA). Differences between the mean values of different groups were analyzed using unpaired t-test.

3. Results

3.1. Hexane extracts of *Annona muricata* and *Piper longum* showed acaricidal properties

The plants *Curculigo orchioides*, *Bauhinia variegata*, *Azadirachta indica*, *Annona muricata*, and *Piper longum* were selected to evaluate their acaricidal properties against *Rhipicephalus microplus* larvae and adults. Soxhlet extraction of 100g of plant materials was carried out using organic solvents, including hexane, chloroform, and methanol. The parts used and the corresponding yield obtained are presented in Table 1. For the Larval Immersion Test (LIT), hexane extracts at 1.25 %, and chloroform and methanol extract at 5 % concentrations were tested. Among

the 15 extracts evaluated, the hexane and chloroform extracts of *A. muricata* and the hexane extract of *P. longum* exhibited the highest larvicidal activity (Fig. 1a). The IC_{50} values of these active plant extracts were further evaluated using LIT. The hexane extract of *A. muricata* demonstrated an IC_{50} value of 0.34 %, while the hexane extract of *P. longum* showed an IC_{50} value of 0.184 % (Fig. 1b and c).

For the Adult Immersion Test (AIT), 5 % concentration of hexane, chloroform, and methanol extracts were used. Significant acaricidal activity was observed with the hexane extracts of *A. muricata* and *P. longum* (Fig. 1d). The IC_{50} values of these active plant extracts were further evaluated. The hexane extracts of *A. muricata* demonstrated an IC_{50} value of 1.39 %, and the hexane extract of *P. longum* showed an IC_{50} value of 0.71 % (Fig. 1e and f).

3.2. Phytochemical screening and fingerprint analysis of active plant extracts

Preliminary phytochemical screening revealed the presence of amino acids, phenols, steroids, and terpenes in the hexane extract of *A. muricata*. Similarly, glycosides, phenols, steroids, and terpenes were identified in the hexane extract of *P. longum* (Table 2). Fingerprint analysis of the active extracts was conducted using High-Performance Thin-Layer Chromatography (HPTLC) for three different batches. The HPTLC analysis identified piperine at R_f 0.35 in the hexane extract of

Table 2

Phytochemical screening of *Annona muricata* and *Piper longum* hexane extract.

S. No	Test	<i>A. muricata</i>	<i>P. longum</i>
1	Carbohydrates (Molisch's)	-	-
2	Amino acids (Ninhydrin)	+	-
3	Steroids & Terpenes (Liebermann-Burchard)	+	+
4	Phenols (Ferric chloride)	+	+
5	Glycosides (Keller - Killiani)	-	+
6	Saponins (Foam test)	-	-

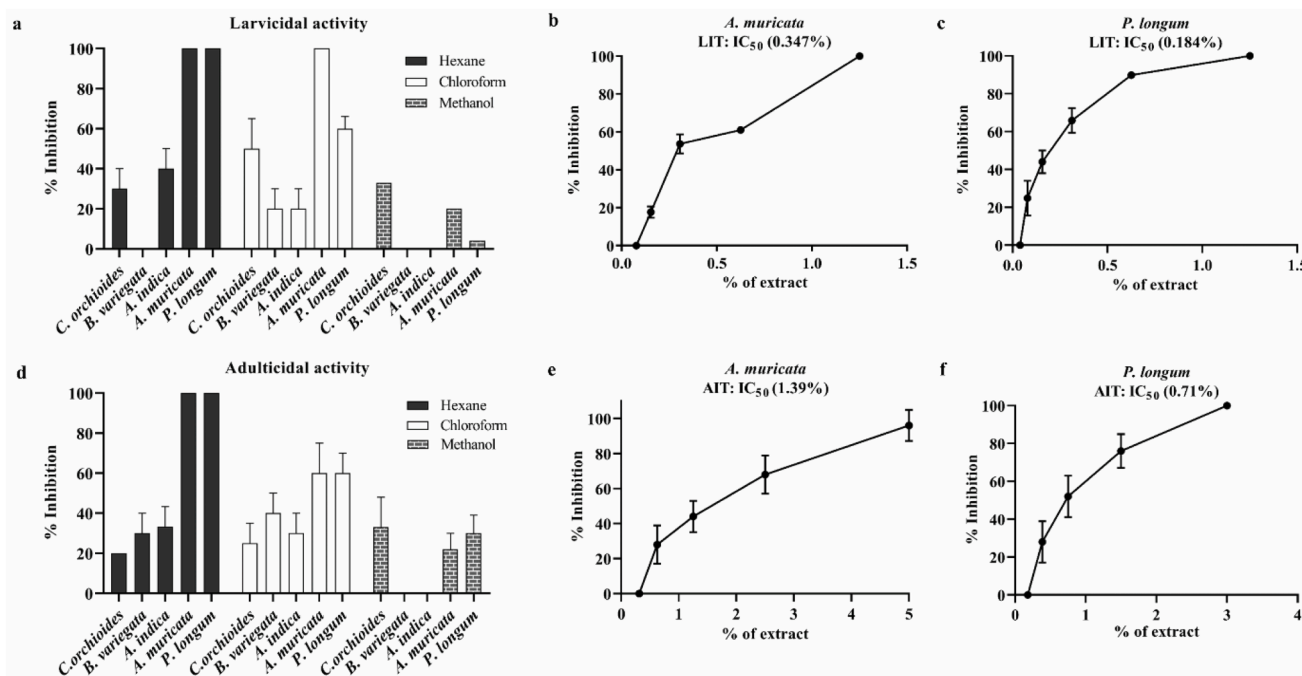


Fig. 1. Effect of plant extracts on *Rhipicephalus microplus* larvae and adults. a. The effect of five plant extracts (hexane, chloroform, and methanol) on larvae was evaluated using the Larval Immersion Test (LIT) for 24 h. The IC_{50} of plant extracts was calculated using LIT for b. *Annona muricata*, and c. *Piper longum*. d. The effect of five plant extracts (hexane, chloroform, and methanol) on adults was evaluated using the Adult Immersion Test (AIT) for 48 h. The IC_{50} of plant extracts was calculated using AIT for e. *Annona muricata*, and f. *Piper longum*. The X-axis represents the percentage of inhibition, and the Y-axis shows the respective extract concentrations used, with IC_{50} values displayed as an insert in the graphs.

P. longum, while α -caryophyllene was identified at Rf 0.58 in the hexane extract of *A. muricata* (Fig. 2a and b). These compounds were further used as marker components to ensure batch-to-batch consistency and quality control of the extracts.

3.3. Evaluation of phytoformulation physical properties and stability

The hexane extracts of *A. muricata* and *P. longum* were mixed with a shampoo base to prepare a phytoformulation. The phytoformulation was tested for various physical properties, including color, odor, pH, solid content (%), foam stability, surface tension (dyne/cm), and wettability (Table 3). The physical properties of the phytoformulation were monitored over 12 months to assess stability (Table 4). Initially, the color of the phytoformulation was green, and it gradually changed to dark green. The pH showed a slight decrease, ranging from 5.9 to 5.7. The surface tension decreased from 33 dyne/cm to 29 dyne/cm. Despite these changes, the odor remained aromatic, foam stability was consistent, and the wettability remained stable at 3s, throughout the observation period.

3.4. Phytoformulations showed good efficacy against deltamethrin-resistant *Rhipicephalus microplus*

Rhipicephalus microplus larvae were tested against synthetic acaricides (Ivermectin, Deltamethrin, and Coumaphos) to assess their susceptibility. Among the acaricides screened, ticks exhibited resistance to deltamethrin. To further evaluate this resistance, the IC_{50} value of deltamethrin was determined to be 78 ppm (Fig. 3a). The resistance ratio (RR) was then calculated by comparing the IC_{50} of the field-collected ticks with a reference susceptible IVRI line I as a reference (11 ppm). The analysis revealed a resistance ratio of seven, indicating level II of resistance in the tick population to deltamethrin.

Further, we evaluated the activity of glutathione S-transferases (GST) and acetylcholinesterase (AChE), two enzymes known for their

Table 3

Physicochemical properties of the phytoformulation.

S. No	Parameters	Formulation
1	Color	Green
2	Odor	Aromatic
3	pH	5.9
4	Solids contents (%)	18
5	Foam stability	Stable
6	Surface tension measurement (dy./cm)	33 $\pm$ 0.5
7	Wettability (s)	3

Table 4

Stability study of herbal phytoformulation physical properties for 12 months.

S. No	Parameters	Months			
		1st	4th	6th	12th
1	Color	Green	Green	Green	Dark Green
2	Odour	Aromatic	Aromatic	Aromatic	Aromatic
3	pH	5.9	5.8	5.8	5.7
4	Solids contents (%)	18	18	18	18
5	Foam stability (mL)	Stable	Stable	Stable	Stable
6	Surface tension measurement (dy. /cm)	33 $\pm$ 0.5	30 $\pm$ 0.5	30 $\pm$ 0.5	29 $\pm$ 0.5
7	Wettability (s)	3	3	3	3
8	Larval Immersion Test (%)	100	100	100	100

critical role in detoxification processes. The GST and AChE activity was measured in tick larvae after treatment with deltamethrin at its IC_{50} . The results showed an increase in AChE activity in treated ticks (0.762 ± 0.04) compared to untreated ticks (0.414 ± 0.04). Similarly, GST activity was higher in treated ticks (0.932 ± 0.1) compared to control ticks (0.659 ± 0.06) (Fig. 3b and c).

The prepared formulation was evaluated through *in vitro* screening against *R. microplus* ticks resistant to deltamethrin. The results demonstrated that the formulation achieved 100 % mortality in both the larval and adult stages of the resistant ticks. Further analysis determined the IC_{50} value of the formulation to be 0.0047 % (concentration of *A. muricata* and *P. longum*=0.00007 %) against larval stage and 30.8 % (concentration of *A. muricata* and *P. longum*=0.046 %) against adult stage (Fig. 3d and e).

3.5. Phytoformulations induced deformities in ovarian and egg structure

Under *in vitro* conditions, at the adult stage, the formulation caused 100 % mortality in ticks after 48 h post-treatment compared to the control group. When the formulation was diluted to a lower concentration, mortality was still observed. The ticks treated with IC_{50} value of the phytoformulation were dissected and examined under a microscope, the significant changes were observed in the ovarian structure compared to the control group (Fig. 4a and b). The hatching of the eggs was observed for 12 days for the ticks treated with IC_{25} value of the phytoformulation. These eggs remained unhatched. Detailed observations using a microscope (Fig. 4c and d) and scanning electron microscope (Fig. 4e and f) revealed structural deformities in the eggs.

3.6. Phytoformulation showed no toxicity on rabbit skin

A toxicity study was conducted to evaluate the safety of the phytoformulation when applied to the skin of rabbits. The rabbits were closely observed for 24 h following the application. During the first six hours, observations for any signs of allergic reactions were made every hour, followed by checks every six hours up to the study period (Fig. S1). The results showed no signs of redness, swelling, irritation, or other allergic reactions on the skin of the rabbits.

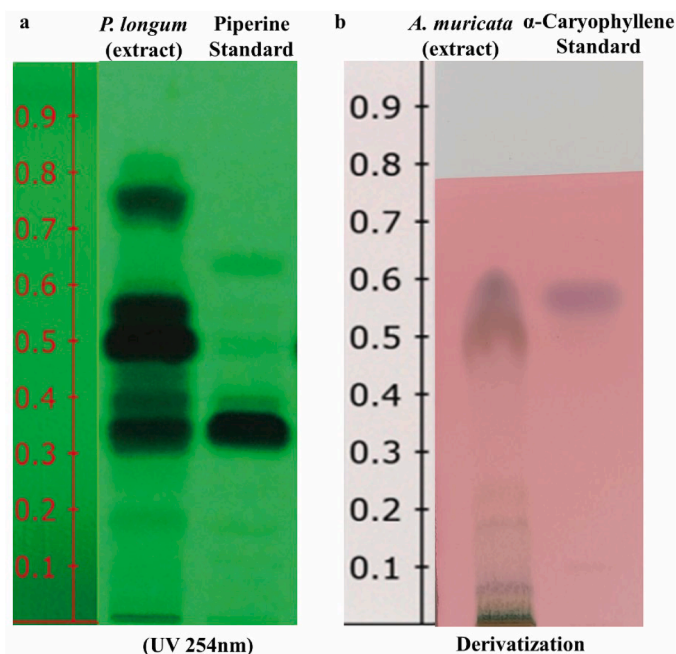


Fig. 2. Active ingredients in hexane extract of *Piper longum* and *Annona muricata*. a. HPTLC profile of the hexane extracts of *P. longum* using a mobile phase of Toluene: Ethyl acetate: Diethyl ether (60:30:10) under UV light 254 nm, with piperine as the active marker. b. HPTLC profile of the hexane extracts of *A. muricata* using a mobile phase of hexane: ethyl acetate (98:2), derivatized with p-anisaldehyde. α -Caryophyllene was used as the active marker.

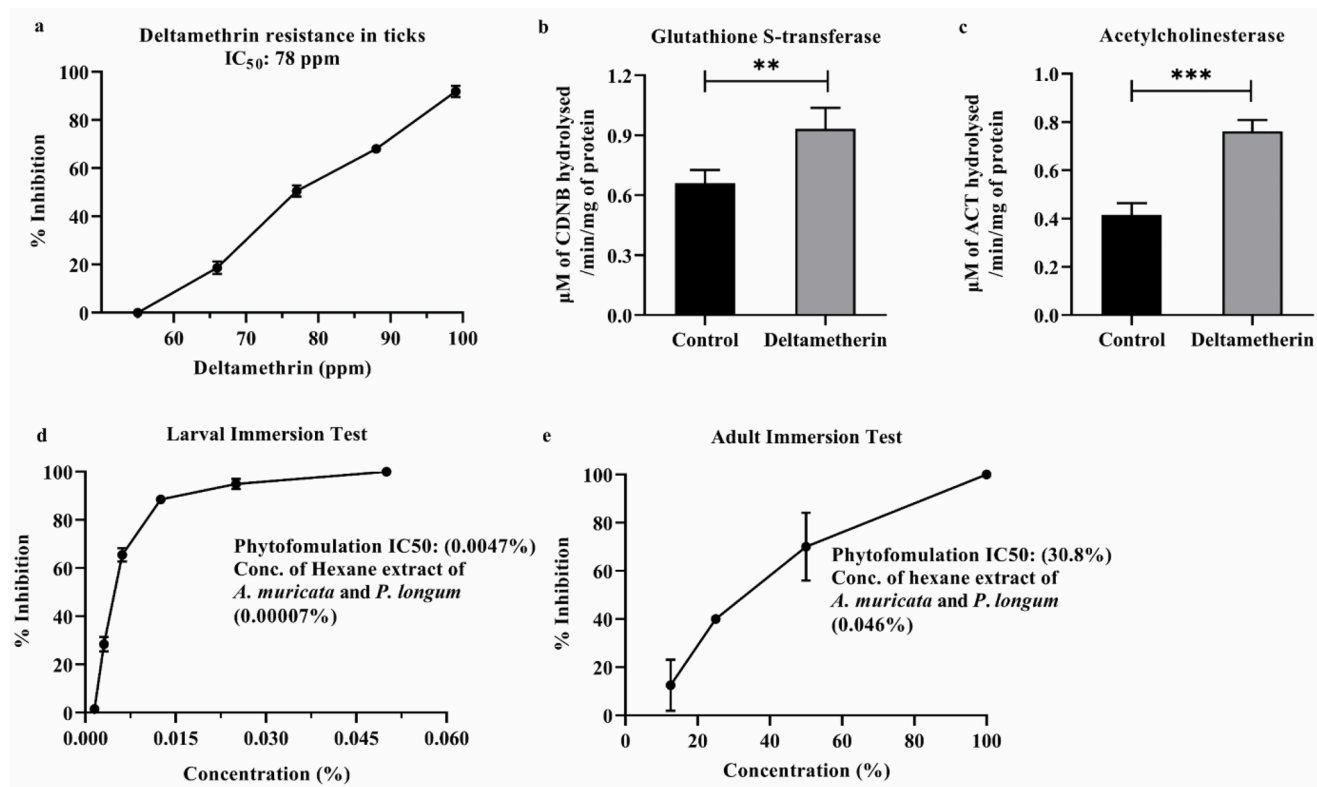


Fig. 3. Assessment of deltamethrin resistance in *Rhipicephalus microplus*, and the efficacy of the phytoformulation using LIT and AIT. a. Dose-response curve of *R. microplus* larvae treated with different concentrations of deltamethrin, used to calculate the IC_{50} value. Quantitative analysis of b. glutathione S-transferase and c. acetylcholinesterase activity in deltamethrin treated and untreated ticks. Evaluation of the phytoformulation's IC_{50} on deltamethrin-resistant d. larvae and e. adults. The IC_{50} and amount of extract in the phytoformulation is represented as an insert in the respective graphs. (***, $p < 0.0005$, **, $p < 0.005$).

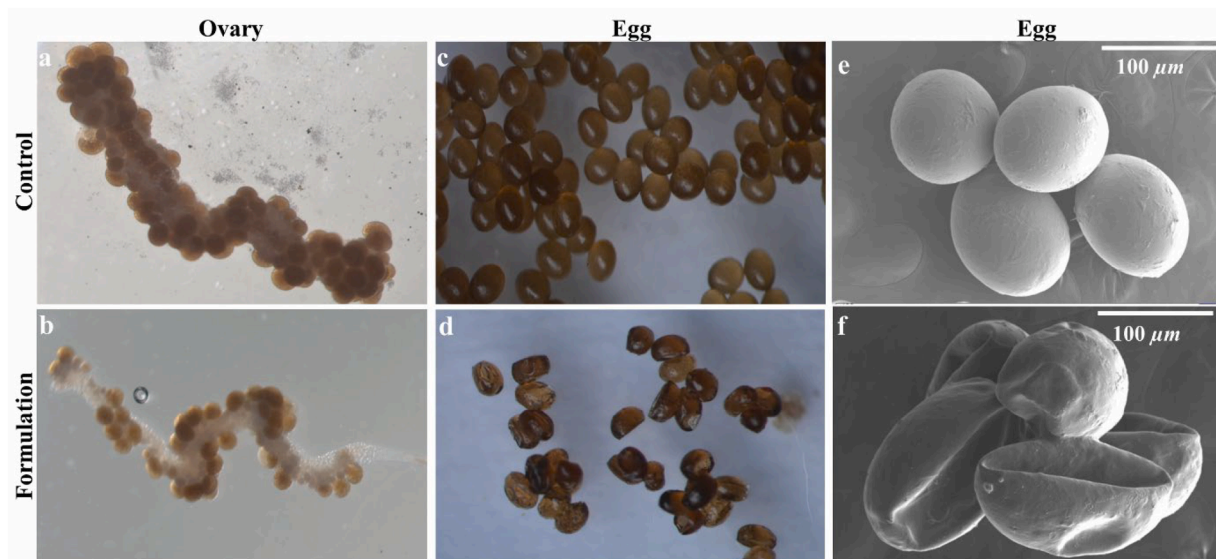


Fig. 4. Phytoformulation treatment impairs ovary structure and egg hatchability in adult *Rhipicephalus microplus* ticks. a. Ovary from control ticks. b. Ovary from ticks treated with IC_{50} value of phytoformulation. c. Eggs from control ticks. d. Eggs from ticks treated with IC_{25} value of formulation. e. SEM image of eggs from control ticks. f. SEM image of eggs from ticks treated with IC_{25} value of formulation. Ovary structures were observed under a stereo microscope at $10\times$ magnification, and egg images were captured using SEM with a $100\ \mu m$ scale.

3.7. Phytoformulation showed promising results in field trials

Field trials of the phytoformulation conducted on naturally tick-infested cattle demonstrated a mean efficacy of $83.3 \pm 3.5\%$ within 48 h of application (Table 5 and Fig. 5). No effect of control (shampoo

base) was observed (data not shown). Before applying phytoformulation on the field animals, a few adult engorged female ticks were collected. The eggs from these ticks were allowed to hatch. Few larvae from the batch were tested for the deltamethrin resistance. All tick larvae were found to be resistant to deltamethrin.

Table 5

Phytoformulation efficacy on naturally infested animals with *Rhipicephalus microplus*.

S. No	Area of application	Treated group I	Treated group II	Treated group III	Treated group IV	Treated group V
1	Neck	85 ± 3.6	85 ± 3.7	82 ± 2	79 ± 2	84 ± 2.5
2	Thigh	85 ± 5	86.5 ± 3.2	79 ± 3.4	81 ± 24	78 ± 6
3	Ears	87 ± 4.5	88 ± 4	83 ± 4	80 ± 3.5	84 ± 3.2
	Average efficacy	86.4 ± 4.5	87 ± 3.5	81 ± 3	80.3 ± 2.6	82.2 ± 4
	Overall efficacy	83.3 ± 3.5				

4. Discussion

The excessive use of chemical acaricides has significantly reduced their efficacy due to the development of acaricide-resistant tick populations. This resistance poses a critical challenge in controlling ticks (Shakya et al., 2020). Hence, it is crucial to discover and develop new acaricides that are more effective against resistant tick populations and have a longer-lasting impact to address the growing issue of acaricide resistance. Plants are one of the most promising sources of alternative acaricides. Many plants possess natural acaricidal properties attributed to secondary metabolites such as alkaloids, flavonoids, terpenoids, and phenolic compounds.

Previous studies have demonstrated that hexane extract of *A. indica* seed significantly impairs the reproductive parameters of engorged female *R. microplus* ticks (Giglioti et al., 2011). Also, recent evaluations of *A. indica* leaf extract in combination with *Eucalyptus* hybrid confirmed significant acaricidal action and oviposition inhibition (Kumar et al., 2023). Taking this clue, we also tested the soxhlation extract of *A. indica* leaf. However, the leaf extracts of *A. indica* did not show any effect on *R. microplus*. Similarly, the aqueous leaf extract of *Curculigo orchoides* adversely affected the development, lifespan, reproduction, and locomotion of *Drosophila melanogaster* (Kushalan et al., 2022). In this study, we evaluated the anti-tick activity of the plant's rhizomes; however, it did not show any activity. The green synthesis of silver nanoparticles using *Bauhinia variegata* leaf extract demonstrated potent larvicidal

activity against *Anopheles subpictus*, *Aedes albopictus*, and *Culex tritaeniorhynchus* (Govindarajan et al., 2016). However, the extracts of the bark of *B. variegata* did not show any acaricidal effect on *R. microplus*. These observations suggest that the specific parts of the three plants used in this study may lack anti-tick activity. In contrast, *Annona muricata* extracts have shown efficacy against mosquito vectors (Parthiban et al., 2020, Santhosh et al., 2015) and, notably, the hexane leaf extract used in this study displayed a clear anti-tick activity at both larval and adult stages. Previous studies have also highlighted the efficacy of methanol extracts of *Piper longum* fruit in affecting tick mortality and reproduction (Godara et al., 2018). Consistent with this, the hexane extract of *P. longum* showed anti-tick activity, with a lower IC₅₀ than that reported for methanol extracts, indicating potentially greater potency. The observed lower IC₅₀ could be due to the use of solvents with varying polarity (lower to higher) in this study. Due to the difference in the polarity of solvents, different secondary metabolites would differentially solubilize, leading to better separation.

It is a common observation that plant extracts often show inconsistent results, largely due to batch-to-batch variations in their phytochemical composition. This inherent variability poses a significant challenge to the reproducibility and standardization of plant-based bioassays. To address this issue, previous studies have emphasized the importance of identifying biomarkers in herbal formulations using HPTLC, enabling reliable quality control and ensuring extract consistency across different batches (Trivedi and Mishra, 2010, Patra et al., 2010, Benelli et al., 2015)). In line with these approaches, HPTLC profiling was carried out for the hexane extracts of *Annona muricata* and *Piper longum*, leading to the identification of the biomarkers, α -carophyllene and piperine, respectively.

Furthermore, the synergistic interactions between natural compounds and synthetic acaricides, as well as among plant-based extracts, are well documented (Teixeira et al., 2023, Gonzaga et al., 2022, Secchi et al., 2022). A combination of *Cymbopogon citratus* and *Eucalyptus globulus* essential oils exhibited enhanced adulticidal activity against *Aedes aegypti*, *A. albopictus*, and *Musca domestica* (Soonwera and Sittichok, 2020). Similarly, a multi-herbal formulation containing extracts from *A. citratus*, *C. citratus*, *Ocimum sanctum*, *Pongamia longifolia*, *Calotropis procera*, *Datura stramonium*, *Aegle marmelos*, *Ricinus communis*,

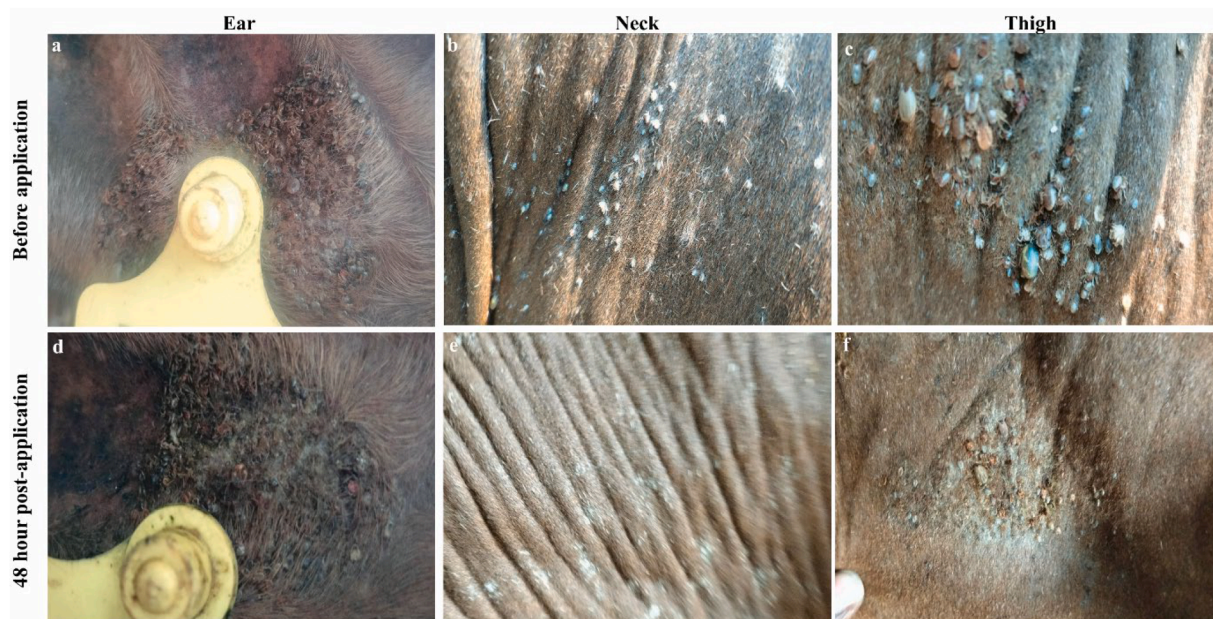


Fig. 5. *In vivo* evaluation of formulation efficacy against naturally infested *Rhipicephalus microplus* ticks on host cattle. Before application of the phytoformulation, a. Ear region, b. Neck region, and c. Thigh region before applying the formulation. After 48 h post-application of the phytoformulation, d. Ear region, e. Neck region, and f. Thigh region. Number of ticks are counted in a 6×6 cm patch on the neck, thigh, and ear areas for each animal.

Azadirachta indica, *Allium sativum*, *Carica papaya*, *Annona squamosa*, and *Piper glabra* significantly reduced tick infestations in cattle and improved hematological parameters (Bhikane et al., 2018). Based on this rationale, we developed a phytoformulation combining hexane extracts of *Piper longum* and *Annona muricata*. To evaluate the performance of the phytoformulation against acaricide-resistant ticks, we classified ticks collected from the field for their acaricide resistance ability by comparing previously reported acaricide resistance data of the reference strains (Sharma et al., 2012). We experimentally observed that ticks were resistant to deltamethrin, consistent with the prior reports of deltamethrin resistance in the area where the samples were collected (Khating et al., 2024). Larval immersion assay using the phytoformulation showed lower IC₅₀ compared to the individual plant extracts in deltamethrin-resistant ticks. Similarly, at the adult stage, the phytoformulation not only kill ticks but also inhibit egg hatchability and causes disruption of ovarian structures. This suggests that the plant extract show synergistic effect, enhancing the overall efficacy of the phytoformulation. In addition, it is effective against deltamethrin-resistant ticks, showing superior acaricidal activity compared to individual plant extracts.

Two key considerations in the development of natural formulations are stability and safety. In our study, the phytoformulation remained stable and retained its acaricidal efficacy for up to 12 months at room temperature. This performance is notably better than previous reports, such as the *Annona squamosa*-based phytoformulation, which showed stability only up to 6 months under various storage conditions before a decline in activity was observed (Sharma et al., 2022). Safety studies on formulation should follow the OECD guidelines. A similar study involving *Cymbopogon citratus* essential oil assessed its toxicity profile in mice and rabbits, confirming its safety in terms of skin irritation and acute/subacute toxicity (Lulekal et al., 2019). In this study, the phytoformulation did not cause any signs of irritation, inflammation, or edema on the application site in albino rabbits. No behavioral changes were observed for up to 24 h post-application, supporting its safety for future use.

A common limitation of plant-based acaricides is the reduction in efficacy under *in vivo* conditions. For instance, a previous study reported that the efficacy of *Acorus calamus* extract was 44 % when tested *in vivo* (Ghosh et al., 2011). In another study, a formulation prepared with 10 % *Annona squamosa* leaves extract showed an efficacy of 77 % in the field study (Sharma et al., 2022). In this study, the phytoformulation containing 3 % of active extracts showed a significant anti-tick activity of 83.3 ± 3.5 % on naturally infested cattle. These findings suggest that the combination of hexane extracts from *Annona muricata* leaves and *Piper longum* seeds maintains consistent efficacy, even against deltamethrin-resistant ticks. This sustained effectiveness could be due to the choice of solvent, the physicochemical properties of the active compounds, and the overall stability of the formulation. However, further evaluation through large-scale field trials across different tick species and controlled infestations is essential to validate these results.

5. Conclusion

This study demonstrates the efficacy of hexane extracts from *A. muricata* leaves and *P. longum* seeds as plant-based acaricides against *R. microplus*. Phytochemical profiling identified α -caryophyllene and piperine as key marker compounds in the active extracts of *A. muricata* and *P. longum*, respectively. A shampoo formulated with these extracts remained stable for 12 months, achieved 100 % mortality *in vitro*, showed no skin toxicity in rabbits, and demonstrated 83.3 ± 3.5 % efficacy in field trials within 48 h. These results suggest the potential of plant-derived acaricides as eco-friendly alternatives to combat acaricide resistance. Further field studies are needed to confirm and optimize these findings for broader applications.

CRedit authorship contribution statement

Macha Vijay: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mood Rajitha:** Writing – review & editing, Methodology, Formal analysis, Data curation. **M Krishnagaanth:** Writing – review & editing, Methodology. **Nitin D. Jadhav:** Resources. **Gajanan Chigure:** Writing – review & editing, Resources, Investigation. **Anand Srivastava:** Writing – review & editing, Visualization, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Macha Vijay reports financial support was provided by Indian Council of Medical Research. Krishnagaanth M reports financial support was provided by India Ministry of Science & Technology Department of Biotechnology. Mood Rajitha reports financial support was provided by Council for Scientific and Industrial Research. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

A.S. thanks the National Institute of Animal Biotechnology (NIAB), Hyderabad, for the core grant supporting this research. A.S. thanks NMPB for funding Z.18017/187/CSS/R&D/TL-01/2016-17-NMPB-IVA. M.V. is supported by a fellowship from the Indian Council of Medical Research (ICMR-SRF 45/14/2020-TR/BMS ICMR) and the Regional Centre for Biotechnology (RCB), Faridabad, India, to pursue a PhD. M.R. is supported by a fellowship from the Council of Scientific and Industrial Research (CSIR), India (09/1150(0009)/2019 -EMR-I). K.M. is supported by a fellowship from the Department of Biotechnology (DBT), Ministry of Science and Technology, India (DBT/2021-22/NIAB/1718). We thank Mr. D. Peddi Reddy, NIAB, Hyderabad, for his help with SEM. We thank Anchrom Enterprises for helping in HPTLC, respectively.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.actatropica.2025.107644](https://doi.org/10.1016/j.actatropica.2025.107644).

Data availability

No data was used for the research described in the article.

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